
Natural Language Processing

Lecture-1 : The Basics of NLP

❖ SUMMARY

Why Do We Use Language?

Language is fundamental because we:

1. Communicate with others
2. Think (partly) through language
3. Tell stories
4. Build scientific theories
5. Form friendships and relationships

Why Do We Need NLP?

NLP helps machines process and understand language in order to:

1. **Access knowledge** (search engines, recommender systems)
2. **Communicate** (e.g., machine translation)
3. **Study language itself** (linguistics & cognitive science)

NLP-Powered Products

1. **Search engines:** Google, Baidu
2. **Social media:** Facebook, Instagram, WeChat, Twitter
3. **Voice assistants:** Alexa, Siri, Google Assistant
4. **Machine Translation:** Google Translate

Why Is Language Hard to Model?

Language is complex because it is a **multi-layered semiotic system**.

Semiotic system : set of rules and procedure for making meaning from signs

Sign : A sign is a pair [...] of signifier(form) and signified(meaning)

Definition of Language

Language is:

- A **means of communication**
- A **semiotic system**

It includes multiple structural levels:

1. Phonological structure : patterns of sounds
 2. Morphological structure : internal structure of words
 3. Syntactic structure : grammatical correctness
 4. Semantic structure : interpretation and understanding
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Levels of Language Analysis

- **Sentence-Level Analysis** : Focuses on the internal structure of language.

Phonological & Graphemic Level

- **Phonological**: Sounds and pronunciation (international phonetic alphabet representation)
- **Graphemic**: Written symbols and spelling patterns

Morphological Level

- Word formation using morphemes
- Example: brav+itude, bio+terror-isme

Syntactic Level

- Grammar and sentence structure
- How words relate (e.g., subject–verb–object)

- **Analysis in Context** : Meaning often depends on context beyond the sentence.

Semantic Level

- Literal meaning
- Identifies “who did what to whom”
- Determines speaker, addressee, and message

Linguistic Context

- Meaning derived from surrounding sentences
- Example: pronoun resolution (“it” referring to a Christmas present)

Extra-linguistic Context

- Uses world knowledge, visual cues, or situation
- Example: A photo clarifies whether “him” refers to a person or a cat

Language Analysis Levels Summary Table

Level	Focus	Key Question
Phonological	Sounds	How is it pronounced?
Morphological	Word structure	How is the word built?
Syntactic	Sentence structure	Is the grammar correct?
Semantic	Literal meaning	What does it mean?
Linguistic Context	Surrounding text	What do references mean?
Extra-linguistic Context	Real-world knowledge	What is the external situation?

The 5 Challenges of NLP

1. Productivity

Language allows speakers to produce and understand an **unlimited number of new sentences**, including ones never heard before .

- New words, meanings, and structures constantly appear.
- Utterances : spoken word , statement or sound

Why it's hard for NLP: Models must handle new words and constructions that were not in training data.

2. Ambiguity

Most speech and text can have **multiple interpretations**.

Humans resolve ambiguity using linguistic and extra-linguistic signals.

Ambiguity appears at all levels:

Syntactic Ambiguity A sentence may have multiple grammatical structures.

Ex : they the ate pizza with anchovies

Semantic Ambiguity Individual words have multiple meanings

- **Polysemy:** multiple related words have the same meaning
 - *set, arm, head*
 - “Head of New Zealand is a woman.”
- **Named entity ambiguity:**
 - “Michael Jordan” (basketball player or professor?)
- **Object/Color ambiguity:**
 - “cherry coat”

Pragmatic Ambiguity (Context and intent)

Meaning depends on interpretation beyond grammar (specific situation or world knowledge):

- “Two Soviet ships collide, one dies.” (one dies may refer to a human not one of the ships)
- “Dealers will hear car talk at noon.”

Discourse-Level Ambiguity

Requires conversation history:

Example dialogue:

- “Where can I find a vegetarian restaurant in Paris?”
- “Give me the top ranked ones in the 14th arrondissement.”
- “How far is the closest one?”

Understanding requires remembering previous references.

Why it’s hard for NLP:

Systems must determine the correct meaning among many possibilities.

3. Variability (Language Variation)

Language varies at all levels:

- Phonetic (accent)
- Morphological & lexical (spelling)
- Syntactic
- Semantic

Phonetic Variation

Pronunciation differences across regions.

Example: Rhotic vs. non-rhotic accents

- In Scotland & Ireland → “arm” pronounced with “r”
- In parts of England → “arm” pronounced “ahm”

Impact: Speech systems must recognize both as the same word.

Spelling & Syntactic Variation (User-Generated Content)

Common in social media.

Example:

- Informal spelling: “wsh”, “ki ca?” instead of “qui ça?”

- Highly informal grammar: “lavé rien fé”

Machine translation systems like Microsoft Translator may fail because they map slang to incorrect standard words.

Solution: Text normalization before processing.

What Determines Variation?

- Where? (home, work, restaurant)
- When? (19th century vs. 2022)
- About what? (domain/topic)
- Who is speaking?
- To whom?

Variation depends on:

- Sociology
- Time
- Topic
- Social context
- Geography

4. Diversity

There are ~7000 languages worldwide.

Only ~60% have written forms.

Languages differ in multiple dimensions:

Phonological Diversity (Sounds)

Languages vary in Vowels and Consonants

Impact: Speech systems trained on small sound inventories struggle with large ones

Graphemic Diversity (Writing Systems)

Not all languages use the Latin alphabet.

Impact: NLP must handle very different text structures.

Syntactic Diversity (Word Order)

Languages differ in Subject (S), Verb (V), Object (O) order:

- SVO → English: “Cats eat mice.”
- SOV → Japanese: “Cats mice eat.”

Some languages have:

- Strict word order (English)
- Free word order (Russian)

Morphology & Word Order Freedom

There is an inverse relationship:

- More word order freedom → less word-position info → more word info → richer morphology

Semantic Diversity Languages divide meaning differently.

Example:

Some languages use **one word** for both “hand” and “arm.”

Others use **two separate words**.

Impact: Machine translation must adapt to each language’s semantic structure.

NLP Variation Summary Table

Category	Dimension of Diversity	Example
Phonologic	Number of vowels/consonants	Rotokas (few) vs. Taa (many)

Category	Dimension of Diversity	Example
Graphemic	Writing system type	Latin Alphabet vs. Chinese Characters
Syntactic	Order of S, V, and O	English (SVO) vs. Hindi (SOV)
Morphologic	Information inside the word	English (Low) vs. Turkish (High)
Semantic	Categorization of concepts	One word for "Hand/Arm" vs. Two

5. Sparsity

Language contains an enormous number of possible word combinations.

Even large datasets cannot cover all possibilities.

- **Low frequency word** → appears rarely
- → many zeros in representation
- → **high sparsity**

Problem:

Many valid words or sentences appear rarely or never in training data.

NLP Challenges Summary Table

Challenge	What it means	Key Idea	Example	Why it's hard for NLP
Productivity	Language can generate unlimited new sentences	New words, meanings, structures appear constantly	New slang or unseen sentence	Model must handle data it has never seen before
Ambiguity	Same sentence/word can have multiple meanings	Exists at all levels (syntactic, semantic, pragmatic, discourse)	"I saw the man with a telescope"	System must choose the correct meaning
Variability (Variation)	Language changes depending on context	Depends on speaker, place, time, topic	Accent differences, slang like "wsh"	Model must handle different forms of the same meaning
Diversity	Many languages differ greatly	Differences in sounds, writing, grammar, meaning	English (SVO) vs Japanese (SOV)	Systems must adapt to very different languages
Sparsity	Many words/structures appear rarely	Low frequency → many zeros in data	Rare words in dataset	Model lacks enough data → poor learning

Statistical Description of Language

- Zipf Distribution applies not only to words but also to:
 - sounds
 - syntactic structures
 - named entities
 - Consequence:
 - Many language units have very low frequency
 - Leads to sparsity → a major challenge in NLP
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What is Natural Language Processing (NLP)?

- NLP = handling complexities of natural language to perform tasks
 - Main transformations:
 - Raw text/speech → structured information
 - Raw text/speech → controlled/generated text or speech
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Basic Framework

- Token: basic unit from a vocabulary (indexed $1 \rightarrow V$)
- Document: sequence of words
 - $d = (w_1, w_2, \dots, w_N)$
- Corpus: collection of documents
 - $D = (d_1, d_2, \dots, d_M)$
- Examples: Wikipedia, NYT articles

Token (Types)

- A token can be a word , subword , character or sequence of characters

Document (Forms)

- A document can be a Sentence , Paragraph or Sequence of characters
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Text Segmentation

- Process: splitting raw text into units of interest
 - Two levels:
 - Split into modeling units (sentence, paragraph, etc.)
 - Split into basic units 'tokens' (words, subwords, characters)
 - Two approaches:
 - Linguistic: word/sentence-based
 - Statistical: frequent subwords (word pieces, sentence pieces)
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Tokenization

- **Definition:** splitting raw text into tokens
 - **Can be treated as a character-level task**
 - **Notes:**
 - Usually easy
 - Hard for some languages (e.g., Chinese) and domains (e.g., social media)
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NLP Tasks (Modeling Framework)

- **Sequence Classification:**
 - Input: sequence of tokens (X) → Output: single label (Y)
- **Sequence Labelling:**
 - One label per token
- **Sequence Prediction:**
 - Output is a sequence of tokens
- **Structure Prediction:**
 - Output is structured (graph, tree, etc.)

NLP Tasks

Part-of-Speech (POS) Tagging

- **Goal:** Find the grammatical category of each word.
- **Example:**
 - Input: [My, name, is, Bob, and, I, live, in, NY, !]
 - Output: [PRON, NOUN, VERB, NOUN, CC, PRON, VERB, PREP, NOUN, PUNCT]

Syntactic Parsing

- **Goal:** Extract the syntactic structure of a sentence.
- **Example:** Dependency Parsing → predicts an acyclic directed graph (tree).

Slot-Filling / Intent Detection

- **Intent Detection:** sequence classification → classify user intent.
- **Slot-Filling:** sequence labeling → identify specific parameters in user request.
- **Example:** Can you please play Hello from Adele?
 - Intent: play_music
 - Slots: [O, O, O, O, SONG, O, ARTIST, O]

Semantic Role Labeling (SRL)

- **Goal:** Identify the semantic role of each predicate in a sentence.
- **Answers** who did what to whom, when, where, why, how.

Named Entity Recognition (NER)

- **Goal:** Find named entities in a sentence.
- **Example:**
 - Input: [My, name, is, Bob, and, I, live, in, NY, !]

- **Output:** [O, O, O, PERSON, O, O, O, O, LOCATION, O]

Machine Translation

- **Translate text from one language to another.**
- **Example:**
 - **Input:** My name is Bob and I live in NY!
 - **Output:** Je m'appelle Bob et je vis à New-York!

Question Answering

- **Goal:** Provide direct answer from text or database.
- **Example:**
 - **Input:** How many episodes in season 2 Breaking Bad?
 - **Output:** 13

Task Type	Input (X)	Output (Y)	Key Idea	Examples	Notes
Sequence Classification	Sequence of tokens (sentence/document)	Single label	Classify the whole input once	1. "This movie is amazing" → Positive 2. sentimentEmail → Spam/Not Spam 3. Sports/Politics/Tech Intent detection	One label per input
Sequence Labelling	Sequence of tokens	Sequence of labels (same length as input)	Label every token individually	1. NER: "Barack Obama lives in USA" → Barack (Person), Obama (Person), USA (Location) 2. POS: "She eats apples" → She (Pronoun), eats (Verb), apples (Noun) 3. Slot filling	Output length = input length
Sequence Prediction	Sequence of tokens	Sequence of tokens (can differ in length)	Generate a new sequence from input	1. Machine Translation: "Hello" → "مرحبا" 2. Text Summarization 3. Chatbot response generation 4. SLR	Output sequence may be longer or shorter
Structure Prediction	Sequence of tokens	Structured object (tree, graph, etc.)	Generate structured output	1. Syntax Parsing: Sentence → parse tree 2. Dependency Parsing: Words connected by subject/object relationships	Output is a complex structure, not just text or labels

Question Answering:

- **Extractive QA (short answers) → Sequence Labelling**
- **Generative QA → Sequence Prediction**

Solving NLP Problems

- Each NLP problem is unique → no universal solution.
- Approaches:
 - Symbolic methods
 - Statistical methods
 - Deep learning methods
- NLP Engineering: Apply what works best based on literature or prior work.
- NLP Research: Develop more accurate, cheaper, faster methods.

Brief History of NLP

Symbolic Era (1940–2000)

- Focus: rule-based systems and formal grammars
- Developed linguistic resources: lexicons, ontologies, grammars

Statistical Learning Era (1990–2010)

- Techniques: Statistical learning theory (SVM, Random Forest), Graphical Probabilistic Models (LDA, HMM)
- Focus on developing annotated datasets

Deep Learning Era (2010–Today)

- Techniques: Deep Learning architectures (Transformers)
- Transfer Learning in NLP: word2vec, BERT, CamemBERT, GPT
- Enabled by more compute, larger raw datasets, and open-source DL libraries

Key Milestones

- 1949 – First Machine Translation “Model” (Warren Weaver)
 - Proposed using electronic computers for translation
 - Applied Shannon’s information theory
 - Modeled context to disambiguate word senses
 - Idea of mapping languages to a universal representation
- 1964 – ELIZA
 - First conversational bot

NLP Workflow

1. Define NLP modules to split a research, engineering, or product problem into tasks.
2. Define evaluation metrics for the system and submodules.
3. Collect data to build/train models.

4. Build baseline models: simple, non-trivial performance models.

5. Build better models using symbolic, statistical, or deep learning methods.

❖ **MCQ**

1. Why do we need Natural Language Processing (NLP)?
 - A. To replace human language
 - B. To analyze and process human language computationally
 - C. To eliminate online text
 - D. To simplify grammar rules**Answer: B**
2. Which of the following is NOT a use of language?
 - A. Building scientific theories
 - B. Making friends and relationships
 - C. Communicating
 - D. Compressing digital images**Answer: D**
3. Language is used for thinking in which way?
 - A. Completely independent of thought
 - B. Partly used for thinking
 - C. Only for emotional expression
 - D. Only for storytelling**Answer: B**
4. One major goal of NLP is to:
 - A. Remove linguistic diversity
 - B. Access knowledge through search engines and recommender systems
 - C. Replace cognitive science
 - D. Reduce internet usage**Answer: B**
5. NLP contributes to communication mainly through:
 - A. Grammar correction only
 - B. Translation systems
 - C. Image processing
 - D. Hardware optimization**Answer: B**
6. NLP is also important for which academic fields?
 - A. Astronomy and Chemistry
 - B. Linguistics and Cognitive Sciences
 - C. Civil Engineering
 - D. Mechanical Design**Answer: B**
7. Approximately how many web pages are online?
 - A. 7 billion
 - B. 70 million
 - C. 70 billion
 - D. 700 billion**Answer: C**
8. How many Wikipedia articles exist approximately?
 - A. 5 million
 - B. 55 million
 - C. 550 million

D. 500 thousand

Answer: B

9. Approximately how many tweets are posted per second?

A. 900

B. 9000

C. 90,000

D. 9 million

Answer: B

10. Approximately how many emails are sent per second?

A. 300,000

B. 3 million

C. 30 million

D. 300 million

Answer: B

11. What percentage of emails are spam?

A. 30%

B. 40%

C. 60%

D. 80%

Answer: C

12. How many people use some form of language globally (Jan 2022)?

A. 4.7 billion

B. 7.9 billion

C. 4.2 billion

D. 3 billion

Answer: B

13. Approximately what percentage of the world population were internet users in January 2021?

A. 54%

B. 59%

C. 70%

D. 80%

Answer: B

14. Approximately what percentage of the world population were social media users in January 2021?

A. 54%

B. 59%

C. 65%

D. 75%

Answer: A

15. Which search engine has more than 2 billion users?

A. Google

B. Baidu

C. Yahoo

D. DuckDuckGo

Answer: A

16. Approximately how many users does Baidu have?

A. 70 million

B. 700 million

C. 7 billion

D. 2 billion

Answer: B

17. Which of the following is an example of a voice assistant?

A. Alexa

- B. Facebook
- C. Wikipedia
- D. Baidu

Answer: A

18. Approximately how many users use Google Translate?

- A. 100 million
- B. 300 million
- C. 500 million
- D. 1 billion

Answer: C

19. According to Definition 1, language is:

- A. A mathematical formula
- B. A semiotic system
- C. A programming language
- D. A biological organ

Answer: B

20. A sign consists of:

- A. Sound and grammar
- B. Signifier and signified
- C. Word and sentence
- D. Subject and object

Answer: B

21. According to Definition 2, a sign includes which structures?

- A. Visual and emotional
- B. Digital and analog
- C. Phonological, morphological, syntactic, and semantic
- D. Logical and computational

Answer: C

22. The phonological level focuses on:

- A. Word formation
- B. Sentence grammar
- C. Sounds and pronunciation
- D. Meaning in context

Answer: C

23. The graphemic level focuses on:

- A. Sound patterns
- B. Written symbols and spelling
- C. Sentence meaning
- D. Word order

Answer: B

24. The morphological level studies:

- A. Sentence punctuation
- B. Word-building using morphemes
- C. Pronunciation rules
- D. Context interpretation

Answer: B

25. The syntactic level is mainly concerned with:

- A. Word sounds
- B. Sentence structure and grammar
- C. External context
- D. Image interpretation

Answer: B

26. The semantic level answers the question:

- A. How is it pronounced?
- B. How is the word built?
- C. Is the grammar correct?
- D. Who did what to whom?

Answer: D

27. Linguistic context refers to:

- A. World knowledge
- B. Images
- C. Surrounding text
- D. Sound patterns

Answer: C

28. Extra-linguistic context refers to:

- A. Sentence grammar
- B. Word formation
- C. Real-world knowledge and situation
- D. Pronunciation

Answer: C

29. Which of the following is one of the five main challenges of NLP?

- A. Compression
- B. Productivity
- C. Encryption
- D. Compilation

Answer: B

30. Productivity in language refers to:

- A. Limiting vocabulary size
- B. The ability to produce and understand infinitely many new utterances
- C. Reducing ambiguity
- D. Fixing grammar rules permanently

Answer: B

31. According to Lyons (1977), productivity allows speakers to:

- A. Use only previously heard sentences
- B. Construct and understand indefinitely many new utterances
- C. Avoid introducing new words
- D. Eliminate variation

Answer: B

32. The addition of words like “staycation” and “social distance” illustrates:

- A. Ambiguity
- B. Sparsity
- C. Productivity
- D. Diversity

Answer: C

33. Ambiguity means that linguistic observations:

- A. Have only one interpretation
- B. Are always grammatically incorrect
- C. Can have multiple interpretations
- D. Cannot be analyzed

Answer: C

34. Humans disambiguate using:

- A. Only grammar rules
- B. Only phonology
- C. Linguistic and extra-linguistic signals

D. Random guessing

Answer: C

35. Ambiguity can appear at which levels?

A. Only semantic

B. Only syntactic

C. All levels (phonology, graphemics, morphology, syntax, semantics)

D. Only phonology

Answer: C

36. Syntactic ambiguity relates to:

A. Word meaning

B. Sentence structure interpretation

C. Pronunciation differences

D. Writing systems

Answer: B

37. Polysemy (e.g., “head”, “arm”, “set”) is an example of:

A. Syntactic ambiguity

B. Semantic ambiguity

C. Phonetic variation

D. Morphological richness

Answer: B

38. The name “Michael Jordan” can refer to different individuals. This is an example of:

A. Named entity ambiguity

B. Phonological diversity

C. Graphemic diversity

D. Sparsity

Answer: A

39. “Your cherry coat” illustrates ambiguity between:

A. Object and color

B. Subject and verb

C. Sound and spelling

D. Tense and aspect

Answer: A

40. “Two Soviet ships collide, one dies” is an example of:

A. Phonetic variation

B. Pragmatic ambiguity

C. Morphological diversity

D. Word order freedom

Answer: B

41. Disambiguation in dialogue often requires:

A. Only dictionary definitions

B. Discourse knowledge

C. Removing context

D. Ignoring previous sentences

Answer: B

42. Language variation can occur at which levels?

A. Only phonetic

B. Only lexical

C. Phonetic, morphological, lexical, syntactic, and semantic

D. Only semantic

Answer: C

43. Phonetic variation mainly refers to differences in:

A. Word meaning

- B. Accent and pronunciation
- C. Sentence structure
- D. Writing systems

Answer: B

44. Spelling and syntactic variation are especially common in:

- A. Scientific journals
- B. Legal documents
- C. User-generated content (social media)
- D. Dictionaries

Answer: C

45. Informal spelling like “wsh” instead of standard writing illustrates:

- A. Morphological richness
- B. Orthographic variation
- C. Semantic partition
- D. Word order constraint

Answer: B

46. Why do standard NLP systems struggle with social media text?

- A. Too much punctuation
- B. Clean grammar
- C. They are trained mostly on clean, formal data
- D. Lack of vowels

Answer: C

47. Language variability depends on:

- A. Only geography
- B. Social context, geography, sociology, date, and topic
- C. Only topic
- D. Only historical period

Answer: B

48. Approximately how many languages are spoken worldwide?

- A. 700
- B. 7000
- C. 70,000
- D. 70

Answer: B

49. About what percentage of languages have a written form?

- A. 20%
- B. 40%
- C. 60%
- D. 90%

Answer: C

50. Phonologic diversity refers to differences in:

- A. Writing systems
- B. Sound inventories (vowels and consonants)
- C. Word order
- D. Word meaning

Answer: B

51. Graphemic diversity refers to differences in:

- A. Pronunciation
- B. Sentence structure
- C. Writing systems (e.g., alphabets, logographies)
- D. Vocabulary size

Answer: C

52. Syntactic diversity is often characterized using:

- A. Vowel length
- B. S, V, O order
- C. Number of morphemes
- D. Alphabet size

Answer: B

53. English follows which typical word order?

- A. SOV
- B. VSO
- C. SVO
- D. OSV

Answer: C

54. Japanese is commonly classified as:

- A. SVO
- B. SOV
- C. VSO
- D. OVS

Answer: B

55. Languages with freer word order usually have:

- A. Poor morphology
- B. No morphology
- C. Richer morphology
- D. No verbs

Answer: C

56. In English, meaning changes significantly if word order changes because English has:

- A. Rich morphology
- B. Thin morphology
- C. Logographic writing
- D. No syntax

Answer: B

57. In morphologically rich languages like Russian, information about grammatical roles is often conveyed by:

- A. Word position only
- B. Word endings (suffixes)
- C. Capital letters
- D. Punctuation

Answer: B

58. Semantic diversity refers to:

- A. Differences in pronunciation
- B. Differences in writing systems
- C. How languages partition the semantic space
- D. Differences in accent

Answer: C

59. Some languages use one word for both “hand” and “arm,” while others use two. This illustrates:

- A. Phonetic variation
- B. Morphological sparsity
- C. Semantic diversity
- D. Syntactic ambiguity

Answer: C

60. Why does semantic diversity make machine translation difficult?

- A. Words are always identical across languages
- B. Each language categorizes concepts differently

- C. All languages have the same vocabulary
- D. Translation ignores meaning

Answer: B

61. What does Zipf's Law describe in language?

- A. Equal frequency of all words
- B. Random distribution of words
- C. Few units are very frequent while many are rare
- D. Only syntactic structures frequency

Answer: C

62. Zipf distributions are observed in which of the following?

- A. Only words
- B. Only sounds
- C. Words, sounds, syntactic structures, and named entities
- D. Only named entities

Answer: C

63. What is the main consequence of Zipf distributions in language?

- A. High uniformity
- B. Data redundancy
- C. Sparsity (many low-frequency units)
- D. Deterministic patterns

Answer: C

64. Why is sparsity challenging for NLP?

- A. Too much repeated data
- B. Too many high-frequency words
- C. Many rare units make learning difficult
- D. Lack of vocabulary

Answer: C

65. In a nutshell, NLP aims to:

- A. Translate only
- B. Store text
- C. Handle language complexities to perform tasks
- D. Replace human language

Answer: C

66. NLP transforms raw text/speech into:

- A. Images
- B. Structured information
- C. Numbers only
- D. Audio signals

Answer: B

67. NLP can also transform raw text/speech into:

- A. Random output
- B. Controlled text/speech
- C. Images
- D. Graphs only

Answer: B

68. In this course, the focus is on:

- A. Speech only
- B. Images
- C. Textual data
- D. Video data

Answer: C

69. A token is defined as:

- A. A sentence
- B. A paragraph
- C. A basic unit from a vocabulary
- D. A document

Answer: C

70. Tokens are indexed from:

- A. 0 to N
- B. 1 to V
- C. 1 to N
- D. 0 to V

Answer: B

71. A document is defined as:

- A. A single word
- B. A sequence of words
- C. A collection of corpora
- D. A token

Answer: B

72. A document is represented as:

- A. $d = (d_1, d_2, \dots, d_M)$
- B. $d = (w_1, w_2, \dots, w_N)$
- C. $D = (w_1, w_2, \dots, w_N)$
- D. $w = (d_1, d_2, \dots, d_M)$

Answer: B

73. In a document, w_n refers to:

- A. First word
- B. Last word
- C. N-th word in sequence
- D. Random word

Answer: C

74. A corpus is defined as:

- A. A sequence of words
- B. A collection of tokens
- C. A collection of documents
- D. A single document

Answer: C

75. A corpus is represented as:

- A. $D = (d_1, d_2, \dots, d_M)$
- B. $D = (w_1, w_2, \dots, w_N)$
- C. $d = (d_1, d_2, \dots, d_M)$
- D. $w = (w_1, w_2, \dots, w_N)$

Answer: A

76. Which is an example of a corpus?

- A. A single sentence
- B. Wikipedia
- C. A token
- D. A word

Answer: B

77. Another example of a corpus is:

- A. One paragraph
- B. NYT articles from 2021
- C. A character

D. A token sequence

Answer: B

78. A token can be:

- A. Only a word
- B. Only a character
- C. Word, sub-word, character, or sequence
- D. Only a sentence

Answer: C

79. A sub-word token example is:

- A. Entire sentence
- B. Sequence of 3 characters
- C. Paragraph
- D. Document

Answer: B

80. A document can be:

- A. Only a paragraph
- B. Only a sentence
- C. Sentence, paragraph, or sequence of characters
- D. Only a token

Answer: C

81. Text segmentation is defined as:

- A. Translating text
- B. Splitting raw text into units of interest
- C. Classifying text
- D. Encoding text

Answer: B

82. The input of text segmentation is:

- A. Tokens
- B. Structured data
- C. Raw text (characters)
- D. Labels

Answer: C

83. The output of text segmentation is:

- A. Images
- B. Units of interest
- C. Audio
- D. Graphs

Answer: B

84. First level of segmentation involves:

- A. Words to characters
- B. Raw text to modeling units
- C. Tokens to sentences
- D. Labels to tokens

Answer: B

85. Modeling units include:

- A. Tokens only
- B. Sentence, paragraph, web page
- C. Characters only
- D. Labels

Answer: B

86. Second level of segmentation involves:

- A. Tokens to labels

- B. Modeling units to tokens
- C. Characters to images
- D. Documents to corpora

Answer: B

87. Basic units in segmentation are referred to as:

- A. Documents
- B. Tokens
- C. Corpora
- D. Labels

Answer: B

88. Linguistically informed segmentation includes:

- A. Random splits
- B. Word and sentence segmentation
- C. Statistical models only
- D. Neural networks

Answer: B

89. Statistically informed segmentation includes:

- A. Grammar rules
- B. Word pieces and sentence pieces
- C. Manual segmentation
- D. Parsing trees

Answer: B

90. Tokenization is defined as:

- A. Classifying tokens
- B. Segmenting text into tokens
- C. Generating sentences
- D. Translating words

Answer: B

91. Tokenization can be framed as:

- A. Word-level task only
- B. Character-level task
- C. Sentence-level task
- D. Document-level task

Answer: B

92. Tokenization is generally:

- A. Always complex
- B. Always impossible
- C. Easy for most languages
- D. Only for English

Answer: C

93. Tokenization becomes complex in:

- A. English only
- B. Chinese and social media
- C. Formal text only
- D. Scientific papers

Answer: B

94. In sequence classification:

- A. Y is a sequence
- B. Y is a single label
- C. X is a label
- D. No tokens involved

Answer: B

95. In sequence classification, X is:

- A. A graph
- B. A sequence of tokens
- C. A label
- D. A document collection

Answer: B

96. In sequence labelling:

- A. One label per sequence
- B. One label per token
- C. No labels
- D. Multiple sequences

Answer: B

97. In sequence prediction:

- A. Y is a single label
- B. Y is a sequence of tokens
- C. Y is a graph
- D. Y is empty

Answer: B

98. In structure prediction, Y can be:

- A. Only a word
- B. Only a sentence
- C. Graph, tree, or complex structure
- D. Only a label

Answer: C

99. What is the main goal of Part-of-Speech (POS) tagging?

- A. Translate sentences
- B. Find the grammatical category of each word
- C. Identify named entities
- D. Detect the user intent

Answer: B

100. In the sentence [My, name, is, Bob, and, I, live, in, NY, !], the POS tag for “name” is:

- A. PRON
- B. NOUN
- C. VERB
- D. CC

Answer: B

101. In the same sentence, the POS tag for “and” is:

- A. PREP
- B. VERB
- C. CC
- D. PUNCT

Answer: C

102. What does Syntactic Parsing aim to extract?

- A. Word embeddings
- B. Semantic roles
- C. Syntactic structure of a sentence
- D. User intent

Answer: C

103. Dependency Parsing predicts:

- A. A sequence of labels
- B. A graph or tree representing sentence structure
- C. A translation

D. Semantic roles

Answer: B

104. Intent Detection is a:

- A. Sequence labelling task
- B. Sequence classification task
- C. Machine translation task
- D. Question answering task

Answer: B

105. Slot-Filling is a:

- A. Sequence labelling task
- B. Sequence classification task
- C. Parsing task
- D. POS tagging task

Answer: A

106. In the example “Can you please play Hello from Adele?”, the intent is:

- A. greet_user
- B. play_music
- C. ask_question
- D. identify_entity

Answer: B

107. In the same example, the slot tag for “Hello” is:

- A. O
- B. ARTIST
- C. SONG
- D. VERB

Answer: C

108. Semantic Role Labelling (SRL) predicts:

- A. Grammatical categories of words
- B. Who did what to whom, when, where, why, how
- C. Named entities
- D. User intent

Answer: B

109. Name Entity Recognition (NER) finds:

- A. Grammatical categories
- B. Semantic roles
- C. Named entities in a sentence
- D. Translation output

Answer: C

110. In the sentence [My, name, is, Bob, and, I, live, in, NY, !], the NER tag for “Bob” is:

- A. LOCATION
- B. O
- C. PERSON
- D. ARTIST

Answer: C

111. In the same sentence, the NER tag for “NY” is:

- A. PERSON
- B. LOCATION
- C. O
- D. SONG

Answer: B

112. Machine Translation input-output example:

INPUT: My name is Bob and I live in NY !

OUTPUT: ?

- A. Je m'appelle Bob et je vis à New-York!
- B. My name is Bob
- C. Hello Bob!
- D. NY est Bob

Answer: A

113. Question Answering takes a question as input and outputs:

- A. Translated text
- B. Predicted labels
- C. Answer to the question
- D. Named entities

Answer: C

114. How do we solve an NLP problem?

- A. Use a universal method for all problems
- B. Each problem is unique; use a toolkit of symbolic, statistical, and deep learning methods
- C. Always use machine translation
- D. Only use deep learning

Answer: B

115. NLP Engineering focuses on:

- A. Creating new languages
- B. Finding what works best for a given method based on existing research
- C. Labeling data manually
- D. Generating random sentences

Answer: B

116. NLP Research focuses on:

- A. Finding a better, more accurate, cheaper, or faster method
- B. Translating sentences
- C. Detecting POS only
- D. Labeling entities

Answer: A

117. The symbolic era of NLP focused on:

- A. Deep learning architectures
- B. Rule-based systems and formal grammars
- C. Statistical models like SVM
- D. Transfer learning

Answer: B

118. During the symbolic era (1940-2000), what resources were developed?

- A. Transformers and GPT models
- B. Annotated datasets only
- C. Lexicons, ontologies, and grammars
- D. Neural networks

Answer: C

119. The statistical learning era of NLP (1990-2010) introduced:

- A. Rule-based parsing
- B. Graphical probabilistic models like HMM and LDA
- C. Transformers
- D. Transfer learning with BERT

Answer: B

120. Which methods were commonly used in statistical learning?

- A. SVM, Random Forest, HMM
- B. GPT, BERT
- C. ELIZA

D. Formal grammars

Answer: A

121. What was developed alongside statistical models in 1990-2010?

- A. Lexicons
- B. Annotated datasets
- C. Transfer learning techniques
- D. Rule-based systems

Answer: B

122. Deep learning era in NLP (2010-today) is characterized by:

- A. Rule-based parsing
- B. Transformer architectures
- C. Only SVM and Random Forest
- D. ELIZA-like bots

Answer: B

123. Transfer learning in NLP includes:

- A. HMM and LDA
- B. word2vec, BERT, CamemBERT, GPT
- C. Formal grammars
- D. Shannon's cryptography method

Answer: B

124. Deep learning era benefits from:

- A. Smaller datasets
- B. More compute, larger datasets, open-source libraries
- C. Only rule-based systems
- D. Manual lexicons

Answer: B

125. In 1949, Warren Weaver proposed:

- A. ELIZA chatbot
- B. Using electronic computers for machine translation
- C. Transformer models
- D. Word embeddings

Answer: B

126. Weaver's 1949 memorandum suggested using:

- A. Shannon's information theory to frame translation as a cryptographic problem
- B. Neural networks
- C. Transfer learning
- D. Word2vec embeddings

Answer: A

127. The first conversational bot developed in 1964 was:

- A. BERT
- B. ELIZA
- C. GPT
- D. CamemBERT

Answer: B

128. In the NLP workflow, the first step is to:

- A. Collect data
- B. Build baseline models
- C. Define NLP modules to split the system
- D. Define evaluation metrics

Answer: C

129. The second step in NLP workflow is:

- A. Build better models

- B. Collect data
- C. Define evaluation metrics for the system and submodules
- D. Train models

Answer: C

130. The third step in NLP workflow is:

- A. Build baseline models
- B. Collect data to build/train models
- C. Define evaluation metrics
- D. Split system into modules

Answer: B

131. Baseline models in NLP are:

- A. The simplest models with non-trivial performance
- B. Always deep learning models
- C. Only statistical models
- D. Only symbolic models

Answer: A

132. The final step in NLP workflow is:

- A. Define modules
- B. Build better models using symbolic, statistical, or deep learning methods
- C. Collect data
- D. Define evaluation metrics

Answer: B